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NPS/001/003 – Technical Specification for 11kV, 20kV and 33kV Overhead Line Air Break Switch-Disconnectors

1. Purpose

This document details the requirements for all 11kV, 20kV and 33kV air break switch disconnectors operating on the distribution network of CE Electric UK.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
DSS/011/009	1.0	Specification for Pole Mounted Switch Disconnectors
TS E019/001	1.0	Overhead Line Air Break Switch-disconnectors
NE OD1	2.0	Specification for 11, 20 & 33kV pole Mounted Air Break Disconnectors.
NPS/001/003	1.0	Specification for Overhead Line Air Break Switch-Disconnectors

2. Scope

This specification details the produce range, design, constructional and technical requirements for 11kV, 20kV and 33kV air break switch disconnectors.

3. Technical Requirements

3.1 General Requirements

Air break switch disconnectors shall comply with ENA TS 41-36 Issue 2: 2004, ENA ER G18 Issue 2:2007 and BS EN 62271 – 102: 2002. The disconnector shall comprise of a three phase, rocking head, gang operated load break switch with three insulators per phase. The kit shall include an operating rod and mechanism to allow operation via a hook stick (see clause 3.2.5).

The equipment shall be suitable for operating under the conditions specified and shall withstand the variations of temperature and atmospheric conditions without undue distortion or deterioration.

All apparatus shall be designed to minimise the risk of accidental short circuit due to birds, vermin and wind-borne debris.

3.2 Technical Requirements

3.2.1 Switch Disconnector Category

Switches supplied under this specification shall be in accordance with (Category 0) as defined in ENA ER G18, clause 1.

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3.2.2 Equipment Ratings

The equipment is required for use on three-phase, 50Hz, 11kV, 20kV or 33kV nominal supply systems with the neutral earthed directly, through an impedance or via a Peterson coil earthing system.

Rated Voltage (kV)	12	24	36
Rated Insulation Level (kV-peak value)			
Lightning Impulse Withstand Voltage			
Common Value	110	150	200
Across the isolating distance	125	165	220
1 Minute Power Frequency Wet Withstand Voltage (kV rms Value) Common Value	45	60	80
across the isolating distance	50	66	88
1 Minute Power Frequency Dry Withstand Voltage (kV rms Value) Common Value	50	70	95
Across the isolating distance	55	77	105
Rated Normal Current (A)	630	630	630
3 Second Short-time withstand Current (kA)	12.5	12.5	20
Short Circuit Making withstand current (kA) Independent Manual	10(25Pk)	10 (25Pk)	10 (25Pk)
Rated Load Breaking Current (A)	630	630	630

3.2.3 Cable and Line Charging Currents

The rated cable and line charging current values for air break switch disconnectors shall comply with those specified in table 6.1 of ENA TS 41-36 Issue 2.

3.2.4 Special Service Conditions

Switches shall be designed to operate normally whilst being coated with a minimum of 10mm of ice as required by Clause 6.2 of ENA TS 41-36 Issue 2. It shall be suitable for use in coastal areas with high salt air pollution.

3.2.5 Hook Stick Mechanisms

It shall be possible to mount the operating mechanism for the switch at a fixed height of 4.3m from ground level, irrespective of the actual pole height and switch mounting position. The pole diameter at this height ranges between 250mm and 450mm. The mechanism shall have an automatic latching pin, which is virtually friction free.

The mechanism shall be designed to be suitable for operation from ground level by the use of standard portable insulating rods.

There must be at least two distinct operations to open or close the switch. The lever shall incorporate an anti-reflex device to prevent reverse movement of the operation rod, operating the equipment. It should be designed for ease of operation, under all service conditions, e.g. severe weather.

Additionally the switch mechanism shall incorporate a "Safety lock feature". The safety lock shall consist of an interlock or additional lever that when applied prevents the switch from being moved from the open position to the closed position. The lever or interlock shall incorporate a label that becomes visible from ground level when the safety lock is applied. The label shall be white lettering on a red background with the words "SAFETY LOCK".

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3.2.6 Insulators

The support insulators shall be manufactured in accordance with ENA TS 43-93 and IEC 61109 and shall be of the silicone rubber type. All insulators shall provide minimum creepage levels of 25mm/kV as detailed for IEC 60815 class 3. They may be of either cylindrical post type or the pedestal post type.

The unit insulators and connection points should be suitably positioned and adequately spaced to facilitate connection and disconnection by Live Line Working Techniques.

3.2.7 Mode of operation

Switches shall of the independent manual type and provide the values of rated short-circuit making current as detailed in table 3.2.3 'Equipment Ratings'.

The switch shall be fully compliant with the requirements for independent manual operation units detailed in clause 1.5.7.1 of ENA TS 41-36 Issue 2.

3.2.8 Contacts

The main fixed and moving contacts shall be self-aligning and shall be manufactured from a suitable high conductivity material.

Where spring loaded devices are used to maintain main contact pressure, they shall be manufactured from corrosion resistant materials which are compatible with the contact. Where separate springs are used, these should not form part of the current carrying circuit. Shrouds may be used as necessary to maintain spring action under ice conditions but they must not act as a trap for moisture or inhibit free air circulation.

In addition to the "over-toggle" mechanism, there should be positive stops in both the closed and open positions, which are independent of the contacts.

The minimum isolating distance across an open switch shall be 300mm for 11kV and 430mm for 20 and 33kV and this gap shall be clearly visible from ground level.

3.2.9 Leads

Any flexible leads which form the main current carrying path shall be of multi-strand, high conductivity annealed copper conductors or laminations of high conductivity strip to BS 4608. They shall preferably be covered for long term protection against abrasion and corrosion. Applied coverings shall fully seal the leads to prevent moisture ingress, however it shall not reduce the flexibility of the lead.

The leads, terminations and supporting apparatus shall be so arranged and connected to provide mechanical stress relief at the connections. Leads shall be either supported, or so designed to provide adequate clearance between groups of leads and moving parts of the switch and to result in minimum wear of both the leads and their coverings.

Compression lugs shall be manufactured from high conductivity electrolytic copper and generally be in accordance with ENA TS 43-92.

3.2.10 Connections

The connections shall be double hole tinned copper NEMA lug 41mm centres (Figure 4 of ENA TS 41-16).

3.2.11 Nuts, Bolts and Studs

All nuts and pins shall be locked in position with locknuts, or with lockwashers, or other suitable devices unless the omission is proved for the application.

All bolts, nuts and washers shall be of non-rusting material and where dissimilar materials are in contact, consideration shall be given to any interactive processes between them. Where electrical connections are involved brass bolts, nuts and washers with a phosphor bronze spring washer shall be used.

All bolts or studs shall project through the nut. Projections exceeding four threads shall be avoided.

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The components used in the disconnecter will be of standard design such that no special spanners or tools shall be required.

All mounting and connecting bolts, nuts, washer's etc. to be provided except pole bolts.

3.2.12 Mounting

The switch shall be suitable for mounting on both single and H type poles.

Manufactures shall provide an option for the supply of common mounting steelwork to accommodate the various mounting options listed above.

All ferrous materials shall be hot dip galvanised in accordance with BS EN 1461 after fabrication.

It shall be possible to offset 2 units to either side of the pole.

3.2.13 Labelling

In addition to the rating plate on the operating mechanisms, the manufacturer shall provide a duplicate rating plate for each switch unit. Labelling shall include the information detailed in IEC 62271-102 Table 4 together with the rated short-circuit making current.

3.3 Testing

All units shall have been subjected to the type tests detailed in Clause 6.6 of ENA TS 41-36 and the routine tests in Clause 6.7 of ENA TS 41-36 Issue 2.

Appendix 1 contains self certification conformance declarations and shall be completed by the supplier with one set per variant offered.

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4. References

4.1 External Documentation

Reference	Title
ENA TS 41-36 Issue 2: 2004	Distribution Switchgear for Service up to 36kV (cable and overhead conductor connected)
ENA TS 41-16: 1981	Apparatus Terminations, Conductor Sizes and Associated Fittings (Copper) used in Indoor and Outdoor Substations with Outdoor Equipment
ENA TS 43-92: 1985	Technical Specification for Overhead Line Connectors
ENA TS 43-93: 2004	Technical Specification for Overhead Line Insulators
ENA ER G18 Issue 2: 2007	Operation of Air Break, Isolating Switches - Recommendations
BS 4608: 1970	Specification for copper for electrical purposes. Rolled sheet, strip and foil
BS EN 12163: 1998	Copper and copper alloys. Rod for general purposes
BS EN 1461: 1999	Hot dip galvanized coatings on fabricated iron and steel articles. Specifications and test methods
IEC 60815: 1996	Guide for the Selection of Insulators in Respect of Polluted Conditions-Edition 1.0
IEC 62271 – 102: 2002	Standards for high-voltage switchgear and control gear

4.2 Internal documentation

Reference	Title

5. Definitions

Term	Definition
BS	British Standard
ENA TS	Energy Networks Association Technical Specification
ENA ER	Energy Networks Association Engineering Recommendation

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
S Salkeld	Standards Engineer	S Salkeld	27/12/07

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M Nicholson	Head of System Strategy	M Nicholson	09/01/08

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Line Standards Manager	G Hammel	27/12/07

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	11/01/08

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards Manager	C Holdsworth	27/12/07

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	27/12/07

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Appendix 1 – Self Certification Conformance Declaration.

Overhead Conductor Connected Air-Break Switch-Disconnectors.

CLAUSE BY CLAUSE CONFORMANCE WITH ENATS 41-36, PART 6- OVERHEAD CONDUCTOR CONNECTED AIR-BREAK SWITCH DISCONNECTORS,

Switchgear covered by ENATS 41-36 shall comply with the latest issues of the relevant International and British Standards. ENATS 41-36 is intended to amplify and/or clarify the requirements of those Standards.

This check sheet identifies the clauses in ENATS 41-36 - Part 6 and the clauses of the aforementioned Standards relevant to overhead conductor connected air-break switch-disconnectors. The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

N/A = Clause is not applicable/appropriate to the product

Cs1 = The test conforms fully with the requirements of this clause

Cs2 = The test conforms partially with the requirements of this clause

Cs3 = The test does not conform to the requirements of this clause

Cs4 = Test not performed, but alternative evidence/ technical case offered

Note - One complete set shall be completed for each variant offered.

Manufacturer:

Product Reference:

Ratings:

Name:

Signature:

Date:

Instructions for completion

When Cs1 code is entered no remark is necessary

- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'IEC' or 'ENATS' as appropriate

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Overhead Conductor Connected Air-Break Switch-Disconnectors

IEC 62271-102, IEC 60265:PART 1				ENATS 41-36 Part 6			Remarks
Clause / Sub-clause IEC 62271-102 1	IEC 60265: PART	Requirement	Confor - mance code	ENATS 41-36 - Part 1 Clause / Sub-clause	Requirement	Confor - mance code	
1	1	General		6.1	General		
				6.1	Class M0 (IEC 62271-102)		
				6.1	Class E0 (IEC62271-102)		
				6.1	Ganged		
2	2	Normal and special service conditions		6.2	Normal and special service conditions		
					Class 10 ice		
3		Definitions		6.3	Definitions		
				6.3.2	Switch-disconnector (CatA)		
				6.3.3	Switch-disconnector (CatB)		
4	4	Ratings		6.4	Ratings		
	4.101			6.4.101	Rated mainly active load breaking current		
	4.102			6.4..102	Rated closed-loop breaking current		
	4.103			6.4.103	Rated no-load transformer breaking current		
	4.104			6.4.104	Rated cable-charge breaking current		
	4.105			6.4.105	Rated line-charging breaking current		

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4.106		Rated values of mechanical endurance for disconnectors		6.4.106	Rated values of mechanical endurance for disconnectors		
				6.4.106	Class M0 – 1000 ops		

IEC 62271-102, IEC 60265:PART 1				ENATS 41-36 Part 6			
Clause / Sub-clause IEC 62271-102 1	IEC 60265: PART	Requirement	Conformance code	ENATS 41-36 - Part 1 Clause / Sub-clause	Requirement	Conformance code	Remarks
4.107		Rated values of electrical endurance for earthing switches		6.4.107	Rated values of electrical endurance for earthing switches		
4.101	4.112	Rated short-circuit making current		6.4.107 6.4.112	Class E0 Rated short-circuit making current for air-break switch disconnectors		
				6.4.112 6.4.112	Table 6.2 4.101 of IEC 62271-102 for earthing switches		
	4.113	Rated breaking and making currents for a general purpose switch		6.4.113	Rated breaking and making currents for a general purpose switch		
				6.4.113	Class E1 (IEC 60265)		
5	5	Design and construction		6.5	Design and construction		
5.1	5.1	Requirements for liquids		6.5.1	Requirements for liquids		
5.2	5.2	Requirements for gasses		6.5.2	Requirements for gasses		
5.3	5.3	Earthing		6.5.3	Earthing		
5.4	5.4	Auxiliary and control equipment		6.5.4	Auxiliary and control equipment		

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5.5	5.5	Dependent power operation		6.5.5	Dependent power operation and dependent manual operation		
				6.5.5.1	a) Anti reflex		
				6.5.5.1	b) Tension for close/open Anti reflex		
				6.5.5.1	c) Motor Actuator retrofit for (a) and (b)		

IEC 62271-102, IEC 60265:PART 1				ENATS 41-36 Part 6			Remarks
Clause / Sub-clause IEC 62271-102 1	IEC 60265: PART	Requirement	Conformance code	ENATS 41-36 - Part 1 Clause / Sub-clause	Requirement	Conformance code	
				6.5.5.2	Insulating insert		
				6.5.5.2	“ “ wet test		
				6.5.5.2	“ “ position		
				6.5.5.2	Mechanical strength		
				6.5.5.2	Prevent water accumulation		
				6.5.5.2	Up (ON), down (OFF), l-r (OFF)		
				6.5.5.2	Handle-1200mm c/l, 750 - 1700mm		
				6.5.5.2	Rod/handle material		
				6.5.5.2	Earth connection- 2 x M12		
				6.5.5.2	Mounting		
				6.5.5.2	Over toggle		
5.6	5.6	Stored energy operation		6.5.6	Stored energy operation		
				6.5.6			
5.7	5.7	Independent manual		6.5.7	Independent manual		

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		operation			operation		
				6.5.7	Integral rods to 6.5.5.2		
5.8	5.8	Operation of releases		6.5.8	Operation of releases		
5.9	5.9	Low and high-pressure interlocking and monitoring devices		6.5.9	Low and high-pressure interlocking and monitoring devices		
5.10	5.10	Nameplates		6.5.10	Nameplates and labeling		
				6.5.10	Short circuit making current		
				6.5.10	Open/close indication		
5.11	5.11	Interlocking devices		6.5.11	Interlocking devices and padlocking facilities		
				6.5.11	Locking from ground level		

IEC 62271-102, IEC 60265:PART 1				ENATS 41-36 Part 6			Remarks
Clause / Sub-clause IEC 62271-102 1	IEC 60265: PART	Requirement	Conformance code	ENATS 41-36 - Part 1 Clause / Sub-clause	Requirement	Conformance code	
				6.5.11	Locking of high level actuation		
				6.5.11	Earthing switch interlocks		
				6.5.11	Key interlock (36kV disconnectors)		
5.12	5.12	Position indication		6.5.12	Position indication		
				6.5.12	Only one visible		
5.13	5.13	Degrees of protection by enclosures		6.5.13	Degrees of protection by enclosures		
5.14	5.14	Creepage distances		6.5.14	Creepage distances		
5.15	5.15	Gas and vacuum tightness		6.5.15	Gas and vacuum tightness		
5.16	5.16	Liquid tightness		6.5.16	Liquid tightness		
5.17	5.17	Flammability		6.5.17	Flammability		

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5.18	5.18	EMC		6.5.18	EMC		
5.101		Special requirements for earthing switches					
5.102		Requirements in respect of the isolating distance of disconnectors					
5.103		Mechanical strength					
5.104		Operation of disconnectors, and earthing switches - position of the movable contact system and its indicating and signaling devices					
5.105		Maximum force required for manual operation					
5.106		Dimensional tolerances					

IEC 62271-102, IEC 60265:PART 1				ENATS 41-36 Part 6			Remarks
Clause / Sub-clause IEC 62271-102 1	IEC 60265: PART	Requirement	Confor - mance code	ENATS 41-36 - Part 1 Clause / Sub-clause	Requirement	Confor - mance code	
				6.5.201	Flexible leads and connections		
				6.5.201	Multi-strand / laminations Covered		
				6.5.201	Stress relief at connections		
				6.5.201	Supported (clearance/ ear)		
				6.5.201	Lugs (BS 2874,ENATS43-92)		
				6.5.202	Surface preparation and		

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					coatings		
				6.5.203	Mounting arrangements		
				6.5.203	ENATS 43-95 (12kV)		
				6.5.203	Fig 10 (36kV)		
				6.5.203	Horizontally / vertically		
				6.5.204	Support insulators - BS EN 60168		
				6.5.205	Contacts		
				6.5.205	Self aligning / high conductivity		
6	6	Type tests		6.6	Type tests		
				6.6	Table 6.3		
6.1	6.1	General					
6.2	6.2	Dielectric tests					
6.3	6.3	Radio interference voltage tests					
6.4	6.4	Measurement of resistance of the main circuit					
6.5	6.5	Temperature rise tests					

IEC 62271-102, IEC 60265:PART 1				ENATS 41-36 Part 6			Remarks
Clause / Sub-clause IEC 62271-102 1	IEC 60265: PART	Requirement	Conformance code	ENATS 41-36 - Part 1 Clause / Sub-clause	Requirement	Conformance code	
6.6	6.6	Short-time withstand current and peak withstand current tests					
6.7	6.7	Verification of the degree of protection					
6.8	6.8	Electromagnetic compatibility (EMC) tests					

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	6.101	Making and breaking tests					
6.102	6.102	Mechanical operations test					
6.103	6.103	Operation under severe ice conditions					
6.104		Operation at temperature limits					
6.105		Test to verify the proper functioning of the position –indicating device					
7	7	Routine tests		6.7	Routine tests		
7.1		Dielectric test on the main circuit					
7.2		Dielectric test on auxiliary and control circuits					
7.3		Measurement of the resistance of the main circuit					
7.4		Tightness					
7.5		Design and visual checks					
7.101	7.101	Mechanical operating tests					

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IEC 62271-102, IEC 60265:PART 1				ENATS 41-36 Part 6			Remarks
Clause / Sub-clause IEC 62271-102 1	IEC 60265: PART	Requirement	Confor - mance code	ENATS 41-36 - Part 1 Clause / Sub-clause	Requirement	Confor - mance code	
8	8	Guide to the selection of, disconnectors and earthing switches		6.8	Guide to the selection of switch-disconnectors, disconnectors and earthing switches		
9	9	Information to be given with enquiries, tenders and orders		6.9	Information to be given with enquiries, tenders and orders - schedule 6.1		
10	10	Rules for transport, storage, installation, operation and maintenance		6.10	Rules for transport, storage, installation, operation and maintenance		
11	11	Safety		6.11	Safety		

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Appendix 2 – Self Certification Conformance Declaration

Type Tests for Overhead Conductor Connected Air-break Switch-disconnector.

Manufacturer:

Product reference:

Ratings:

Name:

Signature:

Date:

Instructions for completion:

- Complete a separate table for each variant and rating
- ENA/SAP to complete columns 1 to 4
- Manufacturer to complete columns 5 to 10
- When test report also covers another rating insert 'See ???A unit' in the Remarks column

N.B. All tests on ???A unit unless otherwise stated

Type test reports table based on ENATS 41-36 Table 6.3

*See bottom of table for conformance declaration codes

** I = Independent; M= Manufacturer; ENA= Energy Networks Association

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	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
	Test Requirement	Specification and standards	Rated value	Test req'd Y or N	Conformance *	Test value	Date of test	Test station Report / Cert No	Witness I, M or ENA **	Remarks
1	Dielectric	IEC60694. Sub-clause 6.2, IEC60265. Sub-clause 6.2. IEC62271-102. Sub-clause 6.2. Tables 1a and 1b of ENATS 41-36.								
2	Wet 1 minute Voltage Withstand test on operating rod insulated insert	Sub-clause 6.5.5.2 of ENATS 41-36								
3	Measurement of resistance of circuits	IEC 60694. Sub-clause 6.4, IEC60265. Sub-clause 6.4, IEC 62271-102. Sub-clause 6.4								
4	Temperature Rise -	IEC 60694. Sub-clause 6.5, IEC60265. Sub-clause 6.5, IEC 62271-102. Sub-clause 6.5								
5	Short-time withstand current and peak withstand current tests.(3sec short time).	IEC 60694.Sub-clause 6.6, IEC60265. Sub-clause 6.6, IEC 62271-102. Sub-clause 6.6.								
6	Verification of protection	IEC 60694.Sub-clause 6.7, IEC60265. Sub-								

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1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
Test Requirement	Specification and standards	Rated value	Test req'd Y or N	Conformance *	Test value	Date of test	Test station Report / Cert	Witness I, M or ENA **	Remarks
	Weatherproofing for outdoor equipment. (outdoor IP3XDW minimum) Mechanical impact. (outdoor – 5J)								
7	EMC tests								
8	Short-circuit making and breaking tests - Switch-disconnector (class E1 general purpose switch - IEC 60265. Sub-clause 3.4.103.1).								
9	Mechanical operations - Disconnector / Switch-disconnector - 1000 operations. (Class M0 disconnector - IEC 62271-102 Sub-clause 3.4.101.1). (Class M1 general purpose switch - IEC 60265 Sub-clause 3.4.103.4).								
10	Mechanical operations - Earthing switch – 1000								
11	Operation under severe ice conditions (10mm thickness								
12	Finish								
13	Process Control								

N/A = Clause is not applicable/appropriate to the product

Cs1 = The test conforms fully with the requirements of this clause

Cs2 = The test conforms partially with the requirements of this clause

Cs3 = The test does not conform to the requirements of this clause

Cs4 = Test not performed, but alternative evidence/ technical case offered

Ct1 = Independent witnessed tests

Ct2 = Not fully independent witnessed tests

Ct3 = Self verified tests

Ct4 = Alternative tests / evidence offered

Ct5 = Manufacturer has underwritten that the product meets the functional and performance requirements without further testing.

Ct6 = Not tested